F70LB Life Insurance Mathematics B

Heriot-Watt University

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13 Insurances on Joint Lives

13.1 Introduction

It is common for life insurance policies and annuities to depend on the death or survival of more than one life. For example:

1. A policy which pays a monthly benefit to a spouse or other dependants after the death of the policyholder (a dependant's pension).
2. A policy which pays a lump-sum on the second death of a couple (often to meet inheritance tax liability).

We will confine attention to policies involving two lives but the same approaches can be extended to any number of lives. We assume that policies are sold to a life age x and a life age y, denoted (x) and (y). The basic contract types we will consider are:

- Assurances paying out *on* first death, and annuities payable *until* first death.
- Assurances paying out *on* second death, and annuities payable *until* second death.
- Assurances and annuities whose payment depends on the *order* of deaths.

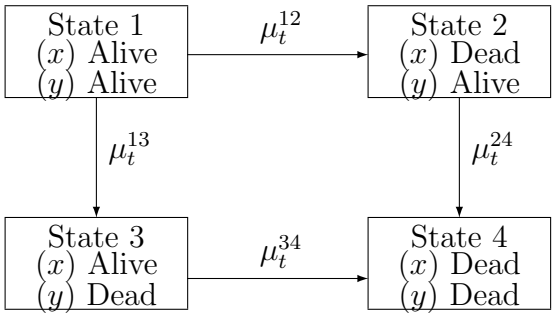
Just as in the single life model, we can define joint life histories using either:

- a multiple-state model and its transition intensities; or
- random variables to represent future lifetimes.

Under certain assumptions these approaches are equivalent. It is simplest to start with the multiple-state model approach.

13.2 Multiple-state model for two lives

The following multiple-state model represents the joint mortality of two lives, (x) and (y).



Notes:

- At time t after policy inception, (x) is age $x+t$ and (y) is age $y+t$, so we just index the transition intensities by time t since policy inception. Other notations are possible.
- We implicitly assume that the simultaneous death of (x) and (y) is impossible: there is no direct transition from state 1 to state 4.

Here we list the main EPVs met in practice, giving their symbols in the standard actuarial notation. For now, we assume

- all **assurance** contracts to be **whole of life**, of **£1** sum assured and payable immediately upon the event that terminates the policy; and
- we assume all **annuity** contracts to be for **whole of life**, **£1** per year and payable continuously until the event that terminates the policy.

13.2.1 Joint-life assurances

- $\bar{A}_{xy}$ is the EPV of £1 paid immediately on the first death of (x) or (y).
- $\bar{A}_{\overline{xy}}$ is the EPV of £1 paid immediately on the second death of (x) and (y).

13.2.2 Contingent assurances

- $\bar{A}^1_{xy}$ is the EPV of £1 paid immediately on the death of (x) *provided* (y) is then alive, i.e. provided (x) is the first to die.
- $\bar{A}_{xy}{}^1$ is the EPV of £1 paid immediately on the death of (y) *provided* (x) is then alive, i.e. provided (y) is the first to die.
- $\bar{A}^2_{xy}$ is the EPV of £1 paid immediately on the death of (x) *provided* (y) is then dead, i.e. provided (x) is the second to die.
- $\bar{A}_{xy}{}^2$ is the EPV of £1 paid immediately on the death of (y) *provided* (x) is then dead, i.e. provided (y) is the second to die.

13.2.3 Joint-life annuities

- $\bar{a}_{xy}$ is the EPV of an annuity of £1 per annum, payable continuously, until the first of (x) or (y) dies.
- $\bar{a}_{\overline{xy}}$ is the EPV of an annuity of £1 per annum, payable continuously, until the second of (x) or (y) dies.

13.2.4 Reversionary annuities

- $\bar{a}_{x|y}$ is the EPV of an annuity of £1 per annum, payable continuously, to (y) as long as (y) is alive and (x) is dead.
- $\bar{a}_{y|x}$ is the EPV of an annuity of £1 per annum, payable continuously, to (x) as long as (x) is alive and (y) is dead.

The notation for reversionary annuities helpfully suggests (taking $\bar{a}_{x|y}$ as an example) an annuity payable to (y) *deferred* until (x) is dead. Think of ${}_m|\bar{a}_{\overline{n}|}$ from Financial Mathematics.

13.2.5 Limited terms

The above contracts may all be written for a limited term of n years, in which case the usual $\overline{n}|$ is appended to the subscript. For example, $\bar{a}_{xy:\overline{n}|}$

13.2.6 Statuses and notation

A status is a condition that continues until it is ended by a specific event.
For example, "(x) is alive" is a status that is ended by the event of the death of (x).
The subscripts xy, $\overline{xy}$ and so on above are called compound statuses; they denote conditions that are ended by some combination of more elementary events, possibly including their order of occurrence.
Where it aids clarity, the elementary statuses may be separated by colons, for example $\bar{A}_{xy}$ is the same as $\bar{A}_{x:y}$, and $\bar{A}_{\overline{xy}}$ is the same as $\bar{A}_{\overline{x:y}}$.

When the event that ends a status happens, the status is said to fail, until then it is said to hold. Note that $\overline{n}|$ defines a deterministic status.

13.2.7 Evaluation of EPVs

In the multiple-state, continuous-time model, we can compute all the above EPVs simply by appropriate choices of assurance benefits b_{ij} or annuity benefits b_i in Thiele's differential equations. The following table lists these choices for whole-life contracts.

EPV	b_1	b_2	b_3	b_{12}	b_{13}	b_{24}	b_{34}
$\bar{A}_{xy}$	0	0	0	1	1	0	0
$\bar{A}_{\overline{xy}}$	0	0	0	0	0	1	1
$\bar{A}^{1}_{xy}$	0	0	0	1	0	0	0
$\bar{A}_{xy}^{\ 1}$	0	0	0	0	1	0	0
$\bar{A}_{xy}^{\ 2}$	0	0	0	0	0	1	0
$\bar{A}^{2}_{xy}$	0	0	0	0	0	0	1
$\bar{a}_{xy}$	1	0	0	0	0	0	0
$\bar{a}_{x\|y}$	0	1	0	0	0	0	0
$\bar{a}_{y\|x}$	0	0	1	0	0	0	0
$\bar{a}_{\overline{xy}}$	1	1	1	0	0	0	0

Example: Consider $\bar{A}_{xy:\overline{n}|}$. Thiele's equations are:

$$\begin{aligned}\frac{d}{dt}V^1(t) &= V^1(t)\,\delta - (\mu_t^{12}+\mu_t^{13})(1-V^1(t))\\ \frac{d}{dt}V^2(t) &= \frac{d}{dt}V^3(t) = \frac{d}{dt}V^4(t) = 0\end{aligned}$$

with boundary conditions $V^1(n) = 1$ and all other $V^i(n) = 0$.

13.2.8 Benefits on Single Lives

This model extends and includes the single-life models for the lifetimes of (x) and (y) separately, as in the following table.

EPV	b_1	b_2	b_3	b_{12}	b_{13}	b_{24}	b_{34}
$\bar{A}_x$	0	0	0	1	0	0	1
$\bar{A}_y$	0	0	0	0	1	1	0
$\bar{a}_x$	1	0	1	0	0	0	0
$\bar{a}_y$	1	1	0	0	0	0	0

13.2.9 Simple Relationships

By considering the cashflows payable while in any state, or on a transition between two states, we can find simple relationships between single-life and joint-life policies. These are useful in calculations. Most common are the following:

EPV	b_{12}	b_{13}	b_{24}	b_{34}
$\bar{A}_x$	1	0	0	1
$\bar{A}_y$	0	1	1	0
$\bar{A}_{xy}$	1	1	0	0
$\bar{A}_{\overline{xy}}$	0	0	1	1

from which we see:

$$\bar{A}_{\overline{xy}} = \bar{A}_x + \bar{A}_y - \bar{A}_{xy}.$$

EPV	b_1	b_2	b_3
$\bar{a}_x$	1	0	1
$\bar{a}_y$	1	1	0
$\bar{a}_{xy}$	1	0	0
$\bar{a}_{\overline{xy}}$	1	1	1

from which we see:

$$\bar{a}_{\overline{xy}} = \bar{a}_x + \bar{a}_y - \bar{a}_{xy}.$$

EPV	b_1	b_2	b_3
$\bar{a}_y$	1	1	0
$\bar{a}_{xy}$	1	0	0
$\bar{a}_{x\|y}$	0	1	0

from which we see:

$$\bar{a}_{x|y} = \bar{a}_y - \bar{a}_{xy}$$

and similarly we can see:

$$\bar{a}_{y|x} = \bar{a}_x - \bar{a}_{xy}.$$

These relationships are important and we summarise them here:

$$\begin{aligned}
\bar{A}_{\overline{xy}} &= \bar{A}_x + \bar{A}_y - \bar{A}_{xy} \\
\bar{a}_{\overline{xy}} &= \bar{a}_x + \bar{a}_y - \bar{a}_{xy} \\
\bar{a}_{x|y} &= \bar{a}_y - \bar{a}_{xy} \\
\bar{a}_{y|x} &= \bar{a}_x - \bar{a}_{xy}
\end{aligned}$$

13.2.10 Composite benefits

Many joint life contracts can be built up out of the above EPVs and the EPVs of single life benefits. This is generally the simplest way to compute them, especially if using tables rather than a spreadsheet. **Example:** Consider a pension of £10,000 per annum, payable continuously as long as (x) is alive, reducing by half on the death of (x) if (x) dies before (y). (This would be a typical retirement pension with a spouse's benefit.) Its EPV, denoted $\bar{a}$ say, is most easily computed by noting that £10,000 p.a. is payable as long as (x) is alive, and in addition, £5,000 p.a. is payable if (y) is alive but (x) is dead, hence:

$$\bar{a} = 10,000\,\bar{a}_x + 5,000\,\bar{a}_{x|y}.$$

13.2.11 Computing contingent assurance EPVs

The one type of joint life benefit whose EPV is not easily written in terms of first-death, second-death and single-life EPVs is the contingent assurance. We defer discussion until Section 13.4.6.

13.2.12 Benefits payable at discrete times

We can also define assurance benefits payable at the end of the year in which the insured event occurs and annuity benefits payable m times per year. We extend our notation in the usual way, so for example:

- A_{xy} is the EPV of a sum assured of £1 payable at the end of the year in which the first of (x) and (y) dies.
- $A_{\overline{xy}:\overline{n}|}$ is the EPV of a sum assured of £1 payable at the end of the year in which the second of (x) and (y) dies, provided this happens within n years, or after n years if at least one of them is then still alive.
- $\ddot{a}_{xy}^{(m)}$ is the EPV of an annuity of £1 per year, payable m times per year in advance, as long as both (x) and (y) are then alive.

Consideration of the tables above (which in fact do not depend at all on whether benefits are payable discretely or continuously) show that useful relationships similar to those in Section 13.2.9 will hold. For example:

$$\begin{aligned} A_{\overline{xy}} &= A_x + A_y - A_{xy} \\ \ddot{a}_{\overline{xy}} &= \ddot{a}_x + \ddot{a}_y - \ddot{a}_{xy} \\ a_{x|y}^{(m)} &= a_y^{(m)} - a_{xy}^{(m)} \end{aligned}$$

13.3 Random joint lifetimes

Now we turn to models defined in terms of random lifetimes. First we note how much of the work we have already done.

13.3.1 Time until failure of a status

Let z denote a status, i.e. a condition that will be ended by a future event, such as the death of a life (x), or the time until failure of a machine. Let T be the time until the event terminating z occurs, assumed to be a random variable with a continuous distribution. We may say that the status z holds until T and fails at T. Then we know all about the machinery to define and calculate EPVs associated with the status z. Define:

$$
\begin{aligned}
F_z(t) &= \text{the distribution function of } T \\
S_z(t) &= \text{the survival function of } T \\
f_z(t) &= \text{the density function of } T \\
\mu_z(t) &= \text{the 'force' associated with } T = f_z(t)/S_z(t) \\
{}_tq_z &= \text{alternative notation for } F_z(t) \\
{}_tp_z &= \text{alternative notation for } S_z(t) \\
K &= \text{the integer part of } T \\
\mathring{e}_z &= \mathrm{E}[T] \\
e_z &= \mathrm{E}[K] \\
\bar{A}_z &= \text{EPV } \pounds 1 \text{ payable at time } T \\
\bar{a}_z &= \text{EPV } \pounds 1 \text{ p.a. continuously until } T \\
A_z &= \text{EPV } \pounds 1 \text{ payable at time } K+1 \\
\ddot{a}_z &= \text{EPV } \pounds 1 \text{ p.a. in advance until } K+1
\end{aligned}
$$

and so on. Then **all** standard results for the single life model carry over with only notational modification: for example:

$$
\begin{aligned}
\bar{A}_z &= \int_0^\infty v^t \; {}_tp_z \; \mu_z(t) \; dt \\
\bar{a}_z &= \int_0^\infty v^t \; {}_tp_z \; dt \\
A_z &= \sum_{k=0}^\infty v^{k+1} \; {}_k|q_z \\
\ddot{a}_z &= \sum_{k=0}^\infty v^k \; {}_kp_z
\end{aligned}
$$

and so on. Therefore, all we need do is work out distributions, densities and forces, given a status z that is ended by the first or second death of (x) and (y); that is, $z = xy$ or $z = \overline{xy}$.

13.3.2 Joint future lifetimes

We have the single life random variables:

$$
\begin{aligned}
T_x &= \text{the future lifetime of } (x) \\
T_y &= \text{the future lifetime of } (y).
\end{aligned}
$$

We will ASSUME that these are independent. This can be a very strong assumption but not much can be done without it in this random lifetime model.
Now define the random variables:

$$
\begin{aligned}
T_{xy} &= \min(T_x, T_y) \text{ and} \\
T_{\overline{xy}} &= \max(T_x, T_y).
\end{aligned}
$$

T_{xy} is the random time until the first death (when status $z = xy$ fails), and $T_{\overline{xy}}$ is the random time until the second death (when status $z = \overline{xy}$ fails). We take as given, from the list above, all 'standard' definitions associated with these statuses:

Note that an alternative notation is sometimes used to indicate explicitly that T_{xy} and $T_{\overline{xy}}$ are the minimum and maximum, respectively, of two random lifetimes:

$$\begin{aligned} T_{\min} &= \min(T_x, T_y) = T_{xy} \text{ and} \\ T_{\max} &= \max(T_x, T_y) = T_{\overline{xy}}. \end{aligned}$$

General z	1st Death $z = xy$	2nd Death $z = \overline{xy}$
$F_z(t)$	$F_{xy}(t)$	$F_{\overline{xy}}(t)$
$S_z(t)$	$S_{xy}(t)$	$S_{\overline{xy}}(t)$
$f_z(t)$	$f_{xy}(t)$	$f_{\overline{xy}}(t)$
$\mu_z(t)$	$\mu_{xy}(t)$	$\mu_{\overline{xy}}(t)$
${}_tq_z$	${}_tq_{xy}$	${}_tq_{\overline{xy}}$
${}_tp_z$	${}_tp_{xy}$	${}_tp_{\overline{xy}}$
K	K_{xy}	$K_{\overline{xy}}$
$\mathring{e}_z$	$\mathring{e}_{xy}$	$\mathring{e}_{\overline{xy}}$
e_z	e_{xy}	$e_{\overline{xy}}$
$\bar{A}_z$	$\bar{A}_{xy}$	$\bar{A}_{\overline{xy}}$
$\bar{a}_z$	$\bar{a}_{xy}$	$\bar{a}_{\overline{xy}}$
A_z	A_{xy}	$A_{\overline{xy}}$
$\ddot{a}_z$	$\ddot{a}_{xy}$	$\ddot{a}_{\overline{xy}}$

and so on.

13.3.3 The distribution of T_{xy}

Because T_x and T_y are assumed to be independent:

$$\begin{aligned} {}_tp_{xy} &= \mathrm{P}[T_{xy} > t] \\ &= \mathrm{P}[T_x > t \text{ and } T_y > t] \\ &= \mathrm{P}[T_x > t] \times \mathrm{P}[T_y > t] \\ &= {}_tp_x \, {}_tp_y. \end{aligned}$$

and (useful result) the joint life survival function ${}_tp_{xy}$ is the product of the single life survival functions. (If T_x and T_y were not independent then we could not write ${}_tp_{xy} = {}_tp_x \, {}_tp_y$.)
Using this result (hence independence):

$$\begin{aligned} {}_tq_{xy} &= 1 - {}_tp_{xy} \\ &= 1 - {}_tp_x \, {}_tp_y \\ &= 1 - \{(1 - {}_tq_x)(1 - {}_tq_y)\} \\ &= {}_tq_x + {}_tq_y - {}_tq_x \, {}_tq_y. \end{aligned}$$

Example: Given ${}_nq_x = 0.2$ and ${}_nq_y = 0.4$ calculate ${}_nq_{xy}$ and ${}_np_{xy}$.
Solution:

$$\begin{aligned} {}_nq_{xy} &= {}_nq_x + {}_nq_y - {}_nq_x\,{}_nq_y \\ &= 0.2 + 0.4 - 0.2 \times 0.4 \\ &= 0.52 \end{aligned}$$

$$\begin{aligned} {}_np_{xy} &= {}_np_x \times {}_np_y \\ &= 0.8 \times 0.6 \\ &= 0.48 \end{aligned}$$

We obtain the density $f_{xy}(t)$ of T_{xy} by differentiation, which again needs the assumed independence:

$$\begin{aligned} f_{xy}(t) &= \frac{d}{dt}\,{}_tq_{xy} = -\frac{d}{dt}\,{}_tp_{xy} = -\frac{d}{dt}\,{}_tp_x\;{}_tp_y \\ &= -\left[{}_tp_x\;\frac{d}{dt}\,{}_tp_y + {}_tp_y\;\frac{d}{dt}\,{}_tp_x\right] \\ &= -\left[{}_tp_x\,(-{}_tp_y\mu_{y+t}) + {}_tp_y\,(-{}_tp_x\mu_{x+t})\right] \\ &= {}_tp_{xy}\,(\mu_{x+t} + \mu_{y+t}). \end{aligned}$$

Compare this to the single life case:

$$f_x(t) = {}_tp_x\,\mu_{x+t}.$$

Hence:

$$\mu_{xy}(t) = f_{xy}(t)/S_{xy}(t) = \mu_{x+t} + \mu_{y+t}$$

(recall that $S_{xy}(t) = {}_tp_{xy}$). This approach uses the assumed independence of T_x and T_y at several points, however the result is true even without independence, as seen in the multiple-state formulation. There, we see that T_{xy} is just the time when state 1 is left. It is an exercise to show that the force associated with this event is $\mu_{x+t} + \mu_{y+t}$ and no independence assumption is needed.

13.3.4 The distribution of $T_{\overline{xy}}$

We have, using the inclusion-exclusion rule:

$$\begin{aligned} {}_tp_{\overline{xy}} &= \mathrm{P}[T_{\overline{xy}} > t)] \\ &= \mathrm{P}[T_x > t \quad \text{or} \quad T_y > t] \\ &= \mathrm{P}[T_x > t] + \mathrm{P}[T_y > t] \\ &\qquad\qquad - \mathrm{P}[T_x > t \text{ and } T_y > t] \\ &= {}_tp_x + {}_tp_y - {}_tp_{xy} \end{aligned}$$

which does *not* require T_x and T_y to be independent. If they are independent then:

$${}_tp_{\overline{xy}} = {}_tp_x + {}_tp_y - {}_tp_x\,{}_tp_y$$

and also:

$$\begin{aligned} {}_tq_{\overline{xy}} &= \mathrm{P}[T_{\overline{xy}} \le t] \\ &= \mathrm{P}[T_x \le t \text{ and } T_y \le t] \\ &= \mathrm{P}[T_x \le t]\mathrm{P}[T_y \le t] = {}_tq_x\,{}_tq_y. \end{aligned}$$

The density $f_{\overline{xy}}(t)$ and the 'force of mortality' $\mu_{\overline{xy}}(t)$ associated with $T_{\overline{xy}}$ are left as tutorial questions.

13.3.5 Expectations of joint lifetimes

The following identities:
Identity 1 $T_{xy} + T_{\overline{xy}} = T_x + T_y$
Identity 2 $T_{xy}\ T_{\overline{xy}} = T_x\ T_y$
can be verified by considering the three exhaustive and exclusive cases:

$$T_x > T_y,\ T_x < T_y \quad \text{and} \quad T_x = T_y.$$

Taking expectations of these gives:

$$\begin{aligned} \mathring{e}_{xy} + \mathring{e}_{\overline{xy}} &= \mathring{e}_x + \mathring{e}_y \\ \mathring{e}_{xy}\ \mathring{e}_{\overline{xy}} &= \mathring{e}_x\ \mathring{e}_y \end{aligned}$$

where in the second of these we have used the assumed independence.
Example: Given:

$${}_tq_x^{\text{non-smoker}} = \frac{70 - x + t}{80 - x}$$

valid for $0 \le x \le 70$ and $0 \le t \le 10$, and that the force of mortality for smokers is twice that for non-smokers, calculate the expected time to first death of a (70) smoker and a (70) non-smoker. You are given that the lives are independent.
Solution: We want:

$$\mathring{e}_{70:70}^{\ s\ \ n-s} = \int_0^{10} {}_tp_{70:70}^{\ s\ \ n-s}\, dt = \int_0^{10} {}_tp_{70}^{s}\, {}_tp_{70}^{n-s}\, dt.$$

Now let μ_x be the force of mortality for non-smokers, then:

$$\begin{aligned} {}_tp_x^s &= \exp\left(-\int_0^t 2\,\mu_{x+r}\, dr\right) \\ &= \left[\exp\left(-\int_0^t \mu_{x+r}\, dr\right)\right]^2 \\ &= \left({}_tp_x^{n-s}\right)^2 \\ &= \left(\frac{10-t}{10}\right)^2. \end{aligned}$$

Since:

$${}_tp_x^{n-s} = 1 - \left(\frac{70 - 70 + t}{80 - 70}\right) = \frac{10-t}{10}$$

therefore:

$$\mathring{e}_{70:70}^{\ s\ \ n-s} = \int_0^{10} \left[\frac{10-t}{10}\right]^3 dt = 2.5 \text{ years}.$$

13.3.6 Evaluation of EPVs

We can evaluate EPVs using the distributions of T_{xy} and $T_{\overline{xy}}$, just as for a single life, as an alternative to solving Thiele's equations. For example, consider an assurance with a sum assured of £ 1 payable immediately on the first death of (x) and (y).

$$\begin{aligned} \text{EPV} = \bar{A}_{xy} &= \mathrm{E}\left[v^{T_{xy}}\right] \\ &= \int_0^\infty v^t\ f_{xy}(t)\ dt \\ &= \int_0^\infty v^t\ {}_tp_{xy}\ \mu_{xy}(t)\ dt. \end{aligned}$$

Note that evaluating an integral numerically amounts to the same thing as solving Thiele's equations numerically.

13.4 Curtate Future Joint Lifetimes

13.4.1 Basic definitions

We have defined:

$$\begin{aligned} K_{xy} &= \text{Integer part of } T_{xy} \\ K_{\overline{xy}} &= \text{Integer part of } T_{\overline{xy}}. \end{aligned}$$

and their respective expected values e_{xy} and $e_{\overline{xy}}$. For more of their properties, see tutorials.
The associated deferred probabilities ${}_{k|}q_{xy}$ and ${}_{k|}q_{\overline{xy}}$ (i.e. the distribution functions of K_{xy} and $K_{\overline{xy}}$ are given by:

$$\begin{aligned} {}_{k|}q_{xy} &= \mathrm{P}[k \le T_{xy} < k+1] \\ &= \mathrm{P}[K_{xy} = k] \\ {}_{k|}q_{\overline{xy}} &= \mathrm{P}[k \le T_{\overline{xy}} < k+1] \\ &= \mathrm{P}[K_{\overline{xy}} = k] \end{aligned}$$

13.4.2 Discrete joint life annuities

Any annuity or assurance we can define as a function of the single lifetime K_x, we can define using K_{xy}, therefore depending on the first death or $K_{\overline{xy}}$, therefore depending on the second death.
Example: Consider an annuity of £1 per annum, payable yearly in advance in advance as long as both (x) and (y) are alive (i.e. payable until the first death). The same reasoning as in the single-life case shows that the present value of this is:

$$\mathrm{PV} = \ddot{a}_{\overline{K_{xy}+1}|}.$$

We denote the EPV of this benefit $\ddot{a}_{xy}$ so:

$$\begin{aligned} \ddot{a}_{xy} &= \mathrm{E}\left[\ddot{a}_{\overline{K_{xy}+1}|}\right] \\ &= \sum_{k=0}^{\infty} \ddot{a}_{\overline{k+1}|}\, {}_{k|}q_{xy} \\ &= \sum_{k=0}^{\infty} v^k\, {}_kp_{xy} \end{aligned}$$

Since the distribution of the present value is known, the variance, $\mathrm{Var}[\ddot{a}_{\overline{K_{xy}+1}|}]$, and other moments can be calculated.
Example: Consider an annuity of £1 per annum, payable yearly in advance in advance as long as at least one of (x) and (y) is alive (i.e. payable until the second death). The same reasoning as before leads to the random present value:

$$\mathrm{PV} = \ddot{a}_{\overline{K_{\overline{xy}}+1}|}.$$

and the EPV:

$$\begin{aligned} \ddot{a}_{\overline{xy}} &= \mathrm{E}\left[\ddot{a}_{\overline{K_{\overline{xy}}+1}|}\right] \\ &= \sum_{k=0}^{\infty} \ddot{a}_{\overline{k+1}|}\, {}_{k|}q_{\overline{xy}} \\ &= \sum_{k=0}^{\infty} v^k\, {}_kp_{\overline{xy}}. \end{aligned}$$

We already know that:

$$\ddot{a}_{\overline{xy}} = \ddot{a}_x + \ddot{a}_y - \ddot{a}_{xy}$$

but we can also see this directly as follows:

$$\begin{aligned}
\ddot{a}_{\overline{xy}} &= \sum_{k=0}^{\infty} v^k \, {}_kp_{\overline{xy}} \\
&= \sum_{k=0}^{\infty} v^k \, ({}_kp_x + {}_kp_y - {}_kp_{xy}) \\
&= \sum_{k=0}^{\infty} v^k \, {}_kp_x + \sum_{k=0}^{\infty} v^k \, {}_kp_y - \sum_{k=0}^{\infty} v^k \, {}_kp_{xy} \\
&= \ddot{a}_x + \ddot{a}_y - \ddot{a}_{xy}.
\end{aligned}$$

13.4.3 Computing $\ddot{a}_{xy}$ etc. using tables

It is easy to compute EPVs of discrete benefits using a spreadsheet if ${}_tp_{xy}$ is known, which is particularly simple if T_x and T_y are independent so ${}_tp_{xy} = {}_tp_x \, {}_tp_y$.

However, other methods can be used if joint life tables are available, and we use these in tutorial questions. Joint life tables usually give values of $\ddot{a}_{xy}$ or a_{xy}, for males age x and females age y. Other quantities can be obtained using these, the single-life tables and simple relationships.

The Orange Tables (2002) use the PMA92C20 and PFA92C20 annuitants' tables and tabulate $\ddot{a}_{xy}$ for selected male ages x and age differences $d = y - x$ with y being female. Linear interpolation is necessary for other ages.

For temporary annuities on joint lives it is usually necessary to go back to first principles, always assuming the independence of T_x and T_y. For example:

$$\begin{aligned}
\ddot{a}_{xy:\overline{n}|} &= \ddot{a}_{xy} - {}_n|\ddot{a}_{xy} \\
&= \ddot{a}_{xy} - v^n \, {}_np_{xy} \, \ddot{a}_{x+n:y+n} \\
&= \ddot{a}_{xy} - v^n \, {}_np_x \, {}_np_y \, \ddot{a}_{x+n:y+n} \\
&= \ddot{a}_{xy} - v^n \frac{l_{x+n}}{l_x} \frac{l_{y+n}}{l_y} \, \ddot{a}_{x+n:y+n}.
\end{aligned}$$

Example: Using PMA/PFA92 mortality and 4% interest calculate the following:

1. $\ddot{a}_{80:75}$
2. ${}_{10|}\ddot{a}_{70:65}$.

Solution:
(1)

$$\ddot{a}_{80:75} = 6.441$$

(2)

$$\begin{aligned}
{}_{10|}\ddot{a}_{70:65} &= v^{10}\, {}_{10}p_{70}\, {}_{10}p_{65}\, \ddot{a}_{80:75} \\
&= v^{10} \left(\frac{l_{80}}{l_{70}}\right) \left(\frac{l_{75}}{l_{65}}\right) \ddot{a}_{80:75} \\
&= (1.04)^{-10} \left(\frac{6953.5}{9238.1}\right) \left(\frac{8784.96}{9703.7}\right) 6.441 \\
&= 2.9651
\end{aligned}$$

We can extend all the varieties of single life annuities, and approximations thereto, to joint life annuities (or indeed to annuities payable while any status z holds). For example:

- Deferred annuities (e.g. previous example)
- Temporary annuities, $\ddot{a}_{xy:\overline{n}|}$ etc.
- Increasing annuities, $(I\ddot{a})_{xy}$ etc.
- Annuities paid monthly. For example:

$$\ddot{a}_{xy}^{(m)} \approx \ddot{a}_{xy} - \frac{m-1}{2m}.$$

- Approximations for annuities paid continuously. For example:

$$\overline{a}_{xy} \approx \ddot{a}_{xy} - 0.5 = a_{xy} + 0.5.$$

Example:
A certain life office issues a last survivor annuity of £2,000 per annum, payable annually in arrear, to a man aged 68 and a woman aged 65.

1. Using the PMA/PFA92 table and a rate of interest of 4% per annum, estimate the expected present value of this benefit.
2. Using the basis of (1) above, derive an expression (which you need NOT evaluate) for the standard deviation of the present value of this benefit, in terms of single life and joint life annuity functions.

Solution:
(1) We want: $2000\, a_{\overline{68:65}}^{m\ \ f}$

$$\begin{aligned}
&= 2000 \left(a_{68}^{m} + a_{65}^{f} - a_{68:65}^{m\ \ f}\right) \\
&= 2000 \left(\ddot{a}_{68}^{m} - 1 + \ddot{a}_{65}^{f} - 1 - (\ddot{a}_{68:65}^{m\ \ f} - 1)\right) \\
&= 2000\ (12.412 + 14.871 - 11.112 - 1) \\
&= 30,342.
\end{aligned}$$

(2) Present Value $= 2000\, a_{\overline{K_{\overline{xy}}|}}$

So we want: var $\left[2000\, a_{\overline{K_{\overline{xy}}|}}\right]$

$$\begin{aligned} &= 2000^2 \text{ var } \left[\ddot{a}_{\overline{K_{\overline{xy}}+1|}} - 1\right] \\ &= 2000^2 \text{ var } \left[\ddot{a}_{\overline{K_{\overline{xy}}+1|}}\right] \\ &= 2000^2 \text{ var } \left[\frac{1 - v^{K_{\overline{xy}}+1}}{d}\right] \end{aligned}$$

$$\begin{aligned} &= \left(\frac{2000}{d}\right)^2 \text{ var } \left[v^{K_{\overline{xy}}+1}\right] \\ &= \left(\frac{2000}{d}\right)^2 \left({}^*A^{\overline{m\ f}}_{68:65} - \left(A^{\overline{m\ f}}_{68:65}\right)^2\right) \end{aligned}$$

where * means evaluated at:

$$j = i^2 + 2i = 8.16\%.$$

We now use:

$$A_{\overline{xy}} = 1 - d\, \ddot{a}_{\overline{xy}}$$

and:

$$\ddot{a}_{\overline{xy}} = \ddot{a}_x + \ddot{a}_y - \ddot{a}_{xy}$$

and:

$$SD[PV] = \sqrt{Var[PV]}$$

to get:

$$\frac{2000}{d} \left\{1 - {}^*d \left({}^*\ddot{a}^m_{68} + {}^*\ddot{a}^f_{65} - {}^*\ddot{a}^{m\ f}_{68:65}\right) - \left[1 - d\left(\ddot{a}^m_{68} + \ddot{a}^f_{65} - \ddot{a}^{m\ f}_{68:65}\right)\right]^2\right\}^{\frac{1}{2}}.$$

13.4.4 Joint life assurances

Consider an assurance with sum assured £1, payable at the end of the year of death of the first of (x) and (y) to die. Its EPV A_{xy} can be written directly:

$$A_{xy} = \text{E}\left[v^{K_{xy}+1}\right] = \sum_{k=0}^{\infty} v^{k+1}\, {}_k|q_{xy}$$

which is easily evaluated using a spreadsheet. If using tables, note that the single-life relationship:

$$A_x = 1 - d\ddot{a}_x$$

extends to any status z:

$$A_z = 1 - d\ddot{a}_z$$

(similarly for continuous benefits). Hence, for example:

$$\begin{aligned} A_{xy} &= 1 - d\ddot{a}_{xy} \\ \bar{A}_{xy} &= 1 - \delta\,\bar{a}_{xy} \\ A_{xy:\overline{n}|} &= 1 - d\ddot{a}_{xy:\overline{n}|} \\ \bar{A}_{xy:\overline{n}|} &= 1 - \delta\,\bar{a}_{xy:\overline{n}|} \end{aligned}$$

and so on.
Associated with the second death we have:

$$A_{\overline{xy}} = \mathrm{E}\left[v^{K_{\overline{xy}}+1}\right] = \sum_{k=0}^{\infty} v^{k+1}\,{}_{k|}\,q_{\overline{xy}}$$

for spreadsheet users, and relationships like:

$$\begin{aligned} A_{\overline{xy}} &= 1 - d\ddot{a}_{\overline{xy}} \\ \bar{A}_{\overline{xy}:\overline{n}|} &= 1 - \delta\,\bar{a}_{\overline{xy}:\overline{n}|} \end{aligned}$$

for users of tables.

13.4.5 Reversionary Annuities

We noted before that:

$$\bar{a}_{x|y} = \bar{a}_y - \bar{a}_{xy}$$

just by noticing that the following define identical cashflows no matter when (x) and (y) should die:

(1) a reversionary annuity of £1 per annum payable continuously while (y) is alive, following the death of (x).

(2) an annuity of £1 per annum payable continuously to (y) for life, *less* an annuity of £1 per annum payable continuously until the first death of (x) and (y).

Hence the cashflows have identical present values, and the same EPVs. We can apply similar reasoning to other variants of reversionary annuities, for example:

$$\begin{aligned} \ddot{a}_{x|y} &= \ddot{a}_y - \ddot{a}_{xy} \\ &= \sum_{k=0}^{\infty} v^k\,{}_kp_y(1 - {}_kp_x) \\ a_{x|y} &= a_y - a_{xy} \\ &= \sum_{k=1}^{\infty} v^k\,{}_kp_y(1 - {}_kp_x) \\ \ddot{a}^{(m)}_{x|y} &= \ddot{a}^{(m)}_y - \ddot{a}^{(m)}_{xy} \\ &= \frac{1}{m}\sum_{k=0}^{\infty} v^{\frac{k}{m}}\,{}_{\frac{k}{m}}p_y(1 - {}_{\frac{k}{m}}p_x) \\ a^{(m)}_{x|y} &= a^{(m)}_y - a^{(m)}_{xy} \\ &= \frac{1}{m}\sum_{k=1}^{\infty} v^{\frac{k}{m}}\,{}_{\frac{k}{m}}p_y(1 - {}_{\frac{k}{m}}p_x). \end{aligned}$$

We can take advantage of some relationships to simplify calculations.

For example:

$$\begin{aligned}\ddot{a}_{x|y} &= \ddot{a}_y - \ddot{a}_{xy} \\ &= (1 + a_y) - (1 + a_{xy}) \\ &= a_y - a_{xy}.\end{aligned}$$

Hence:

$$\ddot{a}_{x|y} = a_{x|y}.$$

Similarly, we can derive the approximate relationships:

1. $\ddot{a}_{x|y} \approx \bar{a}_{x|y}$
2. $\ddot{a}^{(m)}_{x|y} \approx \ddot{a}_{x|y}$.

When using tables, these approximations are generally applied.
Example: Calculate the annual premium (payable while both lives are alive) for the following contract for a man aged 70 and woman aged 64.

Benefits:

- SA of £10,000 payable immediately on first death; and
- a reversionary annuity of £5,000 payable continuously to the survivor for the remainder of their lifetime.

Basis:

- Renewal expenses 10% of all premiums.
- PMA92/PFA92 mortality
- Interest at 4%.

Solution: Let P = Annual Premium.
EPV of premiums less expenses

$$\begin{aligned}&= 0.9\,P\,\ddot{a}^{m\ f}_{70:64} \\ &\approx 0.9\,P\ (10.591) \\ &= 9.532\,P.\end{aligned}$$

where we have needed to interpolate to get $\ddot{a}^{m\ f}_{70:64}$ as follows from the nearest tabulated values:

$$\ddot{a}^{m\ f}_{70:64} \approx \ddot{a}^{m\ f}_{70:60} + \frac{4}{5}(\ddot{a}^{m\ f}_{70:65} - \ddot{a}^{m\ f}_{70:60}) = 10.978 + \frac{4}{5}(10.494 - 10.978) = 10.591$$

EPV of assurance

$$\begin{aligned}&= 10,000\bar{A}^{m\ f}_{70:64} \\ &= 10,000\left(1 - \delta\,\bar{a}^{m\ f}_{70:64}\right) \\ &\approx 10,000\left(1 - \ln 1.04(\ddot{a}^{m\ f}_{70:64} - 0.5)\right) \\ &= 10,000\,(1 - \ln 1.04(10.591 - 0.5)) \\ &= 6042.23\end{aligned}$$

EPV of reversionary annuity

$$\begin{aligned}
&= 5,000\left(\bar{a}^{m\ f}_{70|64} + \bar{a}^{f\ m}_{64|70}\right)\\
&= 5,000\left(a^{m\ f}_{70|64} + a^{f\ m}_{64|70}\right)\\
&\approx 5,000\left(\ddot{a}^{f}_{64} - \ddot{a}^{m\ f}_{70:64} + \ddot{a}^{m}_{70} - \ddot{a}^{f\ m}_{64:70}\right)\\
&\approx 5,000\,(15.242 + 11.562 - 2\times 10.591)\\
&= 28,110.
\end{aligned}$$

Now set: EPV[Income] = EPV[Outgo]

$$\begin{aligned}
\Longrightarrow \quad 9.532\,P &= 6,042 + 28,110\\
\Longrightarrow \quad P &= 3582.90
\end{aligned}$$

13.4.6 Contingent assurances

Contingent assurances are the most awkward to deal with. There are usually no approximations in terms of tabulated functions and they have to be written down as integrals, which have to be evaluated numerically. Note that this is an equivalent problem to solving Thiele's equations.
Consider a contingent assurance with sum assured £1 payable immediately on the death of (x), if (y) is still then alive. The sum assured is payable on the failure of the first-death status $z = xy$, but only if it fails in a certain way:

- If it fails because (x) dies first then the sum assured is paid.
- if it fails because (y) dies first then the sum assured is not paid.

So this benefit does not fit the simple framework of single life, first death and second death assurances (i.e. if the status fails, the benefit is paid). We need to go back to first principles.
First we consider a slightly simpler question: what is the probability of the insured event?
The probability that (x) dies within t years with (y) being alive when life (x) dies is denoted ${}_tq^1_{xy}$ and:

$$\begin{aligned}
{}_tq^1_{xy} &= \mathrm{P}[(x) \text{ dies within } t \text{ years } \mathbf{and} \text{ before } (y)]\\
&= \mathrm{P}[T_x < t \text{ and } T_x < T_y].
\end{aligned}$$

This is an example of a contingent probability, so called because they are associated with the death of a life contingent on the survival or death of another life. To evalauate this we need to consider the full joint probability of T_x and T_y. Define:

$$F_{xy}(r,s) = \mathrm{P}[T_x \le r \text{ and } T_y \le s]$$

and let $f_{xy}(r,s)$ be the associated bivariate density. Assuming independence of T_x and T_y, this density factorises and we have:

$$\begin{aligned}
{}_tq^1_{xy} &= \int_{r=0}^{t}\int_{s=r}^{\infty} f_{xy}(r,s)ds\,dr\\
&= \int_{r=0}^{t}\int_{s=r}^{\infty} f_x(r)f_y(s)ds\,dr\\
&= \int_{r=0}^{t} f_x(r)\left[\int_{s=r}^{\infty} f_y(s)ds\right]dr.
\end{aligned}$$

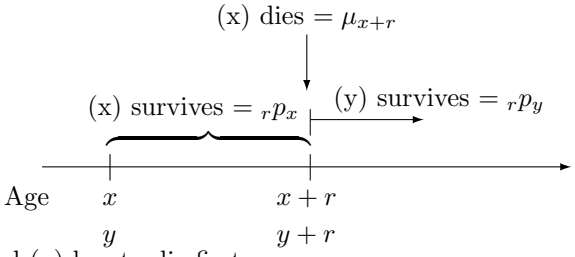
Since:

$$\int_{s=r}^{\infty} f_y(s)ds = {}_rp_y$$

then:

$$ {}_tq^1_{xy} = \int_{r=0}^{t} {}_rp_x \, \mu_{x+r} \, {}_rp_y \, dr. $$

Illustration: (x) survives to time r and then dies and (y) survives beyond r, giving: ${}_rp_x \, \mu_{x+r} \, {}_rp_y$



Note that since one of (x) and (y) has to die first:

$$ {}_tq^1_{xy} + {}_tq_{xy}{}^1 = {}_tq_{xy}. $$

We define ${}_tq^2_{xy}$ as the probability that (x) dies within t years but after (y) has died.
Therefore:

$$ {}_tq_x = {}_tq^1_{xy} + {}_tq^2_{xy} $$

from which:

$$ {}_tq^2_{xy} = \int_0^t {}_rp_x \, \mu_{x+r} \, (1 - {}_rp_y) \, dr. $$

With these probabilities we can now evaluate the EPV $\bar{A}^1_{xy}$.

$$ \bar{A}^1_{xy} = \begin{cases} v^{T_x} & \text{if } T_x < T_y \\ 0 & \text{if } T_x \geq T_y. \end{cases} $$

From this it can be shown that:

$$ \bar{A}^1_{xy} = \int_0^\infty v^r \, {}_rp_x \, \mu_{x+r} \, {}_rp_y \, dr. $$

Similarly, we can derive an expression for the EPV of a benefit of £1 payable immediately on the death of (x) if it is after that of (y), denoted $\bar{A}^2_{xy}$:

$$ \bar{A}^2_{xy} = \int_0^\infty v^r \, {}_rp_x \, \mu_{x+r} \, (1 - {}_rp_y) \, dr. $$

We see, what is intuitively clear, that:

$$ \bar{A}_x = \bar{A}^1_{xy} + \bar{A}^2_{xy}. $$

Other important relationships:

1. $\bar{A}_{xy} = \bar{A}^1_{xy} + \bar{A}_{xy}{}^1$ and $\bar{A}_{\overline{xy}} = \bar{A}^2_{xy} + \bar{A}_{xy}{}^2$.
2. $A^1_{xy} = \sum_{t=0}^{\infty} v^{t+1} {}_tp_x p_y \cdot q^1_{x+t:y+t}$.
3. $A_{xy} = A^1_{xy} + A_{xy}{}^1$ and $A_{\overline{xy}} = A^2_{xy} + A_{xy}{}^2$.
4. $A^1_{xy} \approx (1+i)^{-1/2} \, \bar{A}^1_{xy}$.

Example

1. Express ${}_nq^2_{xy}$ in terms of single life probabilities and contingent probabilities referring to the first death.
2. Suppose $\mu_x = \frac{1}{80-x}$ for $0 \leq x \leq 80$,
   evaluate ${}_{20}q^2_{40:50}$.

Solution
(a) ${}_tq_{xy}^{\ 2}$

$$\begin{aligned}
&= \int_{r=0}^{t} {}_rp_y\,\mu_{y+r}\;(1-{}_rp_x)\;dr \\
&= \int_{r=0}^{t} {}_rp_y\,\mu_{y+r}\,dr - \int_{r=0}^{t} {}_rp_y\,\mu_{y+r}\,{}_rp_x\,dr \\
&= {}_tq_y - {}_tq_{xy}^{\ 1}.
\end{aligned}$$

(b) ${}_nq_x = 1 - {}_np_x$

$$\begin{aligned}
&= 1 - \exp\left\{-\int_0^n \mu_{x+t}\,dt\right\} \\
&= 1 - \exp\left\{-\int_0^n \frac{1}{80-x-t}\,dt\right\} \\
&= 1 - \exp\left\{[\log(80-x-t)]_0^n\right\} \\
&= 1 - \exp\left\{\log\left(\frac{80-x-n}{80-x}\right)\right\} \\
&= \frac{n}{80-x}.
\end{aligned}$$

and we have ${}_nq_{xy}^{\ 1}$

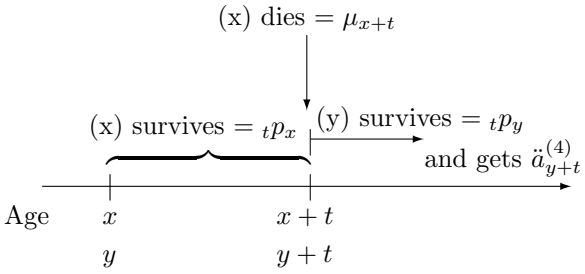
$$\begin{aligned}
&= \int_0^n {}_tp_x\,{}_tp_y\,\mu_{y+t}\,dt \\
&= \frac{1}{(80-x)(80-y)}\int_0^n (80-x-t)\,dt \\
&= \frac{n(80-x)-\frac{1}{2}n^2}{(80-x)(80-y)}
\end{aligned}$$

$$\begin{aligned}
\therefore {}_{20}q_{40:50}^{\ 2} &= {}_{20}q_{50} - {}_{20}q_{40:50}^{\ 1} \\
&= \frac{20}{30} - \frac{20\times 40 - \frac{1}{2}\,20^2}{40\times 30} \\
&= \frac{1}{6}.
\end{aligned}$$

Example
Consider a reversionary annuity of £1 p.a. payable quarterly in advance during the lifetime of (y) following the death of (x). Show that the expected present value of this benefit is approximately equal to:

$$a_{x|y} + \frac{1}{8}\bar{A}^1_{xy}.$$

Solution
Hint: Fix a time t to represent the death of (x), as below:



Hence:

$$\begin{aligned}
\text{EPV} &= \int_0^\infty v^t \, {}_tp_x \, \mu_{x+t} \, {}_tp_y \, \ddot{a}^{(4)}_{y+t} \, dt \\
&\approx \int_0^\infty v^t \, {}_tp_x \, \mu_{x+t} \, {}_tp_y \left(\bar{a}_{y+t} + \frac{1}{8} \right) dt \\
&= \int_0^\infty v^t \, {}_tp_x \, \mu_{x+t} \, {}_tp_y \, \bar{a}_{y+t} \, dt \\
&\qquad + \frac{1}{8} \int_0^\infty v^t \, {}_tp_x \, \mu_{x+t} \, {}_tp_y \, dt \\
&= \bar{a}_{x|y} + \frac{1}{8} \, \bar{A}^1_{xy} \approx a_{x|y} + \frac{1}{8} \, \bar{A}^1_{xy}.
\end{aligned}$$